

## Putting Together Your Strongest K99/R00 Application

The NIH Pathway to Independence Award (K99/R00) is designed to help postdoctoral researchers complete needed, mentored training and transition in a timely manner to independent, tenure-track (or equivalent) faculty positions. The K99/R00 award is intended to foster the development of an independent research program that will be competitive for subsequent independent funding and will help advance the mission of the NIH. Candidates must have [no more than 4 years of postdoctoral research experience](#) at the time of the initial or the subsequent resubmission or revision application. The K99/R00 award is intended for individuals who require at least 12 months of mentored research and career development (K99 phase) before transitioning to the R00 award phase of the program.

The strongest candidates will require, and will propose, a well-conceived plan for 1–2 years of substantive mentored research and career development that will help them become competitive candidates for tenure-track faculty positions and prepare them to launch robust, independent research programs. An individual who cannot provide a compelling rationale for at least one year of additional mentored research experience and career development at the time of award is not a strong candidate for this award.

The challenge for K99 applicants is presenting a complete, consistent, and compelling story of how the proposed research and training plans will build on past accomplishments to achieve the applicant's career goals. The proposal must be synergistic—all components working together to foster a successful transition to independence.

Consider using a “gap analysis” framework to help you identify key training activities to achieve your goals and harmonize information across the different sections of the grant application. One way to approach this is thinking about the “future” (goals and needed skills), “past” (previous experiences and how you’ve developed your skills to-date), and “present” (the “gap,” or growth opportunity between current and future skills and your training plan to address this).

### Before You Start Your Application

1. **Define your career goals.** Explicitly define your career goals and area of research interest. Your research interest should be *specific*: proposing to study “mechanisms of normal and pathological aging” does not help reviewers envision you as an independent researcher. The research topic should be different enough from your current mentor's research to show independent thinking, but similar enough that they will be able to guide you in your journey. Additionally, the research topic should clearly build on your previous research experiences.

2. **Outline the techniques, skills, knowledge, and relationships necessary to achieve your career goals.** Describe the scientific and professional skills you will need as an independent researcher. When you transition to independence, what techniques will your lab use to investigate your area of research interest? What are the skills that you, as an early-career scientist, will need to successfully run a lab? How will you integrate into the community of your field? What funding will be applied for in the near future? What publications will be generated? The [National Postdoctoral Association's Core Competencies](#) is one source of ideas.
3. **Perform a skills “gap analysis.”** What skills from the above description have you already mastered? Where is the growth opportunity? What skills need to be developed? What are the research strengths you have developed so far and how do they provide a foundation for further progress? Exactly how will this K99/R00 award meet your needs and help you achieve independence? Both the career plan and research plan should move you towards your goals.
4. **Define your research plan.** How will this research advance the science and your career? How does it build off your existing strengths to provide you with skills, techniques, and data that will form the foundation of your research program and first R01 (or equivalent) submission? Are your aims hypothesis-driven and independent of each other? What are your contingency plans if the research doesn't go as planned?
5. **Build a career plan that is tailored to your needs.** What activities will you perform to develop your professional skills? Research specific courses, technical training, seminars, and conferences to attend. What skills will you learn from each? Put together a detailed, feasible timeline.
6. **Assess guidance and mentorship needed.** Take stock of your needs and goals to identify mentors and advisors who can guide you during this time, both at your institution and outside it, if appropriate. Do they have a history of successfully mentoring trainees to independent positions? What will you learn from each identified mentor; are all your areas of development covered?
7. **Put together your K99/R00 application. Or choose a different award.** After assessing your career goals and performing a skills gap analysis, you may find that the K99/R00 is not a good fit. You may not be ready to apply, in which case a different award may be more appropriate. Alternatively, you may be ready to start the job search. Please reach out to NINDS to discuss your options.

## **The Application**

K99/R00 applications are reviewed on five criteria: Candidate; Career Development Plan/Career Goals & Objectives; Research Plan; Mentor(s), Co-Mentor(s), Consultant(s), Collaborator(s); and Environment & Institutional Commitment to the Candidate. **Visit the Notice of Funding Opportunity (NOFO) to make sure you address the required review criteria for each section** (see "Section V. Application Review Information" of the NOFO).

Tips for each section are below:

### **Candidate**

- Take pains to make sure your application presents a consistent and compelling case for your potential to develop a creative, independent research program. Your application, including your biosketch; personal statement; letters of support; mentor statement; and institutional support letter should be harmonious in their presentation of your career goals and potential.
- Present a coherent picture of how your current skills and expertise were developed through your pre- and postdoctoral research training experience, and how each research experience shaped your current research interests.
- Demonstrate research productivity. If your publication history is weak, explain any previous gaps and present a specific plan (topics and timeline) for building a sufficient publication record to be competitive for an independent position within about a year of the start of the award. Remember that peer review panels typically recognize preprints as real markers of productivity.
- Choose references that have *current* knowledge of your abilities, activities, and goals and can write you a strong, detailed letter.
- Clearly describe how you plan to scientifically separate from your mentor and advance to research independence. This should be described and should align in both the candidate section and in your mentor's letter.

### **Career Development Plan**

- Present a specific plan that clearly builds on *your* existing strengths and weaknesses and explains how each proposed activity is in service of your career goals. You should demonstrate how, once training is complete, you will be fully equipped to compete for faculty jobs and transition to independence.
- Include both formal activities and research components to build scientific expertise and professional skills. Ensure that the activities/components are tailored to meet your needs and to meet the goal of achieving independence as a researcher.

Examples of activities/components include: technical training, coursework, career development workshops, research seminars, teaching activities, grant opportunities, and scientific conference attendance. Include names of courses (if proposed) and describe percent effort or hours dedicated to each aspect of training. Provide a timeline for the planned activities.

- Provide information on the frequency of meetings with mentors (and how frequently the mentors will meet with each other) and advisory committee, if included.

## **Research Plan**

- Be clear how the research plan enhances your existing research skills and sets you up for independence. Make sure the proposed research is relevant to your stated career objectives.
- The strongest research plans include a clear rationale with hypothesis-driven aims that are independent of each other. Clearly articulate the significance of the research in a way that is understandable to non-experts.
- Fully describe your methods and include a detailed analysis plan including power analyses. Ensure that your application includes components that address [reproducibility, rigor, and transparency](#).
- Be clear about your role in designing the experiments and collecting preliminary data.
- Include contingency plans if the research doesn't go as planned. What are the potential pitfalls and alternate approaches?
- How will you distinguish your work from that of your mentor? Be explicit about what aspects of the project will you have "ownership" of and can take with you to an independent position.
- Clearly define K99 *and* R00 activities. Provide a rationale for the order in which aims will be addressed, and during which phase. Provide evidence of long-term viability of the proposed independent phase research plan, and how the research will position you for a successful R01 (or equivalent) grant submission.

## **Mentoring**

- Your mentor should demonstrate a strong track record in training and transitioning postdocs to independent careers. Mentor(s) should demonstrate productivity, quality of publications, and adequate funding. Your mentor's research qualifications and experience should be appropriate for your specific career development needs.

If your mentor is a new investigator or lacks mentoring experience, consider adding an experienced co-mentor to your application.

- Many applicants form a mentoring team or advisory committee with complementary and relevant expertise. Provide detailed plans for regular interaction with your mentor(s). Letters of support from mentor(s) should be strong and explicitly address the review criteria on which you will be evaluated, including your potential as well as your strengths and areas needing improvement.
- Your mentor(s) should include a comprehensive plan to support your career development and research plans and your efforts to transition to independence. Describe how (and how often), including a timeline with milestones, your mentors will evaluate your progress. Ensure your mentor's statement and letters of support align with what you have written in the rest of the application.

### **Environment and Commitment**

- The institutional commitment letter from the dean or chair should describe how they will *strongly* support you in the K99 phase and help you to transition. The institution should demonstrate that they are committed to you regardless of the outcome of review of this application. There should be an explicit commitment to a minimum of 75% protected time, with specific details of how this was calculated.
- Describe research facilities and educational opportunities, including collaborating faculty.
- Describe any unique features of the scientific environment that benefit the proposed research, e.g., useful collaborative arrangements or subject populations.

### **Other Suggestions**

- Be *specific* in all aspects of your application. Name courses to be taken, fully describe methods and analyses, provide detailed timelines, etc. Don't leave anything vague.
- Explain why *you* are doing this research and these activities. The "why" is important; don't assume reviewers will connect the dots on their own. Make sure that your application is tailored to *you*.
- Make sure all aspects of the application are *consistent* in terms of your career goals, strengths and weaknesses, and the research and training plan (e.g., if you are proposing to learn a certain technique, then you must explain how that technique is necessary to achieve your career goals).

- Avoid typos and grammatical errors. Have a colleague read the entire application for errors.
- Soliciting feedback from scientists who are not familiar with you or your work is the best way to evaluate how compelling your application is. Ask a third party to evaluate whether you have addressed all of the required review criteria (see "Section V. Application Review Information" of the [NOFO](#)).